



KALN

Aluminum Nitride

KEY PROPERTIES

- High thermal conductivity
- Electrical insulator
- Good mechanical properties
- Thermal expansion similar to semi-conductors

APPLICATIONS

- Heat sinks & heat spreaders
- Electronic insulating components
- Substrates for microelectronic devices

MAIN PROPERTIES

Property	Value	Unit
Density	3.30	g/cm ³
Flexural Strength	330	MPa
Porosity	0	%
Thermal Conductivity (100°C)	>165	W/m·K
CTE (20-500°C)	4.5	x10 ⁻⁶ K ⁻¹
Volume Resistivity	1E13	Ω·cm
Dielectric strength	>15	kV/mm

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	1350	°C
Max working temp (inert)	1900	°C
Recomended temp	1500	°C
Continuous temp	1400	°C

ELECTRICAL BEHAVIOR

Dielectric strength	>15
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MATERIAL INFORMATION

Material type	AlN (pure)
Compability	Yes
Structure	Dense ceramic
Purity	High

The information provided is for guidance only and does not constitute a guarantee.

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